

Neuro - Symbolic and Explainable AI Systems

• Hybrid Processing

→ Connectionist and NN

↳ Neural Networks emulate brains plasticity and functional networks

→ Massively parallel processing

→ PLEthora of parameter tuning

→ Function approximations vs cognition

→ Symbolic AI

IF... AND... AND THEN...

↳ Strong AI research in inference systems to model human recognition

→ Symbolic representations and their manipulation; symbolic understanding

→ Sequential list processing, recursion (Prolog, Lisp)

→ Small domains only

• Hybrid Neural Symbolic Processing

→ brain as a hybrid system supporting different forms of processing: signals, symbols, structures, knowledge

→ Hybrid systems in AI and knowledge engineering for increasing performance

→ Hybrid processing in cognitive science and cognitive neuroscience for plausible cognitive models

→ Explainable AI (XAI) can use symbolic representations

Benefit of Hybrid Representation Integration

	Neural / Statistical Sub-symbolic Data Mining	Symbolic / Structural / Rule-Based
Knowledge Formal Representation	Numbers, Connections	Rules, Trees, Structures
Computational Elements	Distributed Numerical Associations Weights Threshold	Localist Premises, Conclusions Rule Strength Predicates
Processing	Continuous Numbers	Discrete Symbols
Cognitive Level	low	high
Basic Units	NN, Statistics	Rules
Manipulated by c.c.	Continuous Math	Symbolic Logic
Representation	Compact but Distributed	Verbose (leading to brittleness)

Eg:

- Cognition is not precise ("good close to the talk")
- Different forms of properties and form of processing
 - multimodal integration of cognitive processing
 - speech / language integration (signals & concepts)
 - information retrieval (text & images)

Motivate

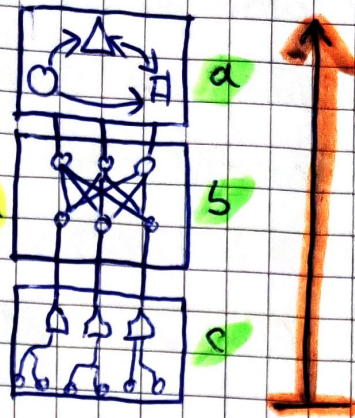
- ↳ Integration of hybrid methods in one system
 - integration of Neural & Symbolic Techniques
 - Combination of symbolic, fuzzy and statistical methods

Neural Symbolic Knowledge Technology Archs

a) Symbolic Knowledge and Understanding

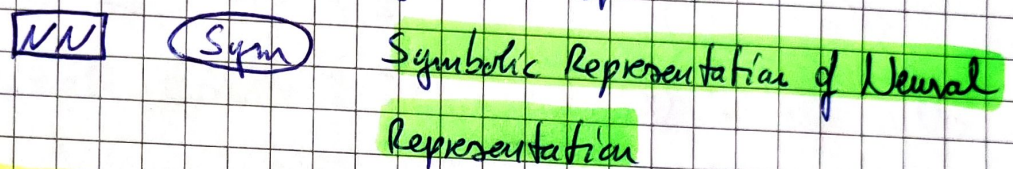
b) Neural / Statistical Knowledge Representation

c) Sensory input from several modalities (audio, vision, etc)



Types of Hybrid Integration Archs

→ Hybrid Transfer Arch: knowledge is transf



→ Hybrid Processing Arch for hybrid realization of symbolic structures

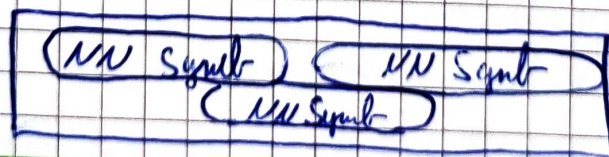
→ Loose Coupling

Symbolic and Neural Modules separate and unidirectional communication

→ Tight Coupling

Symbolic and Neural Modules separate and bidirectional communication

→ Integration



Symbolic and Connectionist Modules fully embedded and integrated (work together)

Hybrid Transformation Archs

→ Symbolic representation of neural representation

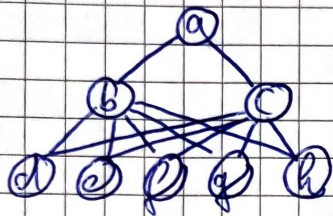
→ Automatic Insertion or Extraction of Symbolic Knowledge

→ Explanation of NN

Weight-Based Transfer

- Connectionist weights contain the knowledge
- Problem: distributed representations are difficult to understand and modify
- Transfer for weights into symbolic rules
- Often used for feedforward NN
- Grouping, Elimination and Clustering of weights; N-of-M rules

eg: Feedforward NN



Symbolic Rules

a: -b, c

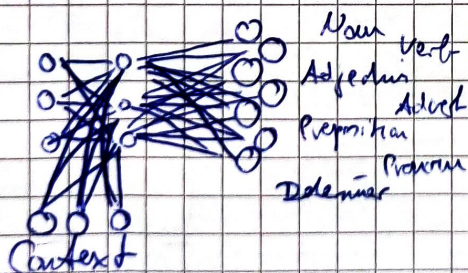
b: -d, e, f

c: -f, g

Knowledge Extraction from Transducer Networks

- recurrent transducer networks for syntactic phrase analysis
- What is trained in eg's and what is to be learned is the mapping between basic and more abstract syntactic categories

Output:
Noun Group
Verb Group
Prepositional Phrase



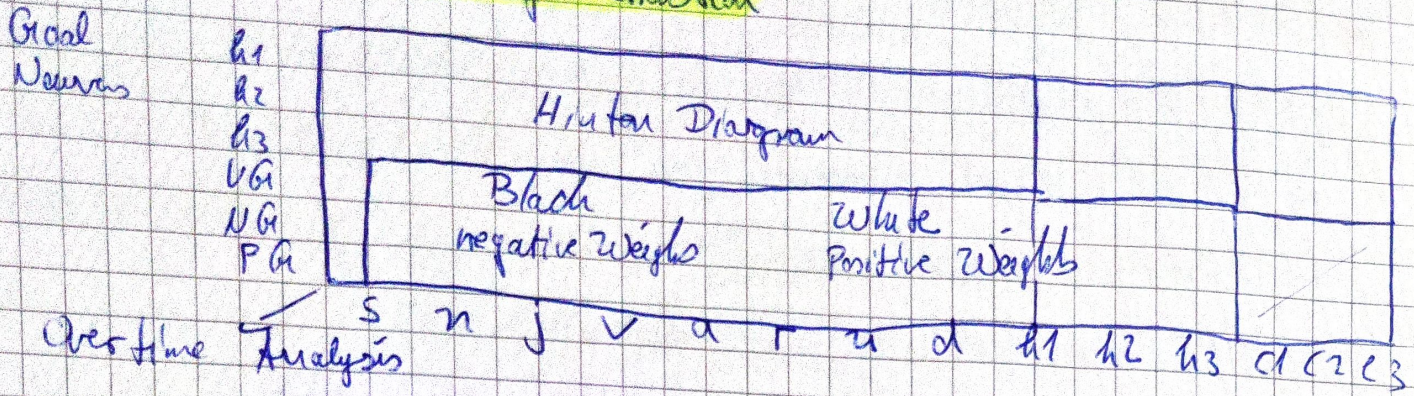
Input

prepositional
↑
I (n → up)
though (v → ug)
in (n → pg) the
(n → pg) next (j → p)
which (n → pg)

How to interpret RNN?

- Cluster analysis, activation values, weight interpretation, eg: Hinton Diagram
- Stepwise Dynamic Analysis of the learning behavior over time
- Symbolic interpretation of NN as transducers (Kruschke mit Zeit)

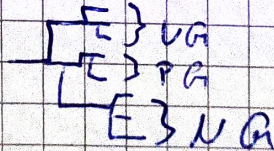
Weight Analysis for Knowledge Extraction



- can explain certain phenomena with weight analysis
- Difficult to directly interpret and extract explicit knowledge
- Static representation of weights does not show dynamics of recurrent networks
- distribution of weights difficult to draw for larger networks

Activation Analysis for Knowledge Extraction

- cluster the activation patterns of all internal representations of sequences
- The internal layer represents the learned knowledge about abstract categories → dendrogram



→ Using PCA

VG
PG / NG

Stepwise dynamic analysis of the learning behavior

- initial high error equally distributed
- learning of often occurring noun groups
- learning of less often prepositional groups, verb groups later
- learning of sequential ~~order~~ context later
- learning of exceptions

• Activation-based automata extraction

- weights represent knowledge at a very detailed level
- Activation values represent more integrated knowledge for particular pattern
- Activations of internal representations can be used for automata extraction
- RNN transferred to finite state automaton

eg: if state is x and input is y then go to state z

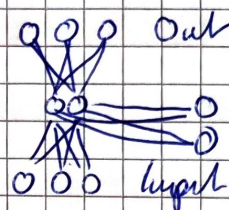
⇒ Preference Moore Machines and RNNs

Output $O = [0, 1]^m$

Preference Mapping

States
 $S = [0, 1]^p$

Input $I = [0, 1]^n$

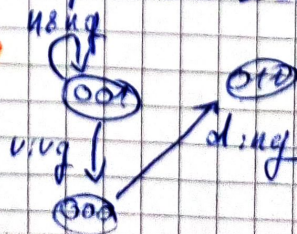


→ If state is a and input is b then go to state x and output is y

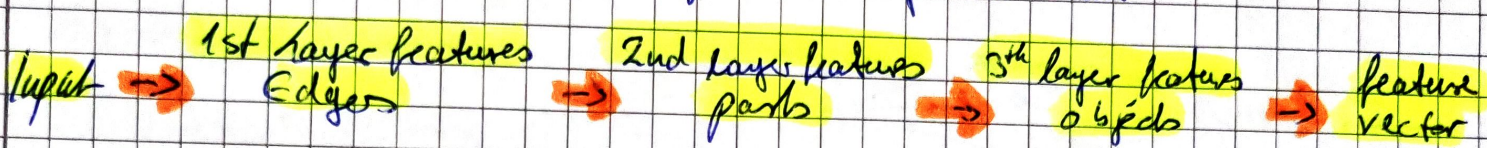
- weights $\hat{=}$ knowledge statically
- Processing is needed to represent knowledge more dynamically
- RNN transferred to finite state automaton
- $\forall$ output vector and $\forall$ Concatenation of input and state vector find the closest symbolic representations (closest ~~corner~~ corner preference vector according to some metric, eg Euclidean)

• Symbolic interpretation of RNN as transducers

- Mapping from vectors to symbolic representation
- Presentation of all patterns
- Computation of next corner preference for each output vector & state vector
- Computation of symbolic transducer
- Flexibility of abstraction level



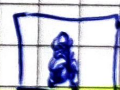
- Towards explainable Deep learning: **Learning hierarchical features**
 - use of trained **feature extractors** to ~~the~~ model data representation in CNN
 - Using sequences of layers to obtain **hierarchical data representation**: from low level features to higher level of abstraction



- each layer increases the level of abstraction of the modelled data
 - Image: pixel → edges → shapes → objects
 - Text: character → word → sentence → story
 - Speech: sound → phone → phoneme → word

• How can we explain what a network "knows"?

→ **Eg:** Which area was used to classify an image



Class Activation Map (CAM)

→ **Class Activation Map (CAM)**

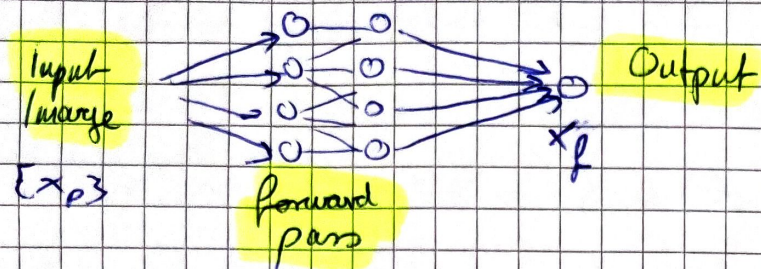
- ↳ visualize relevant input features (what parts were used to classify)
- ↳ regarding a specific class

• Explaining the **internals of RL**

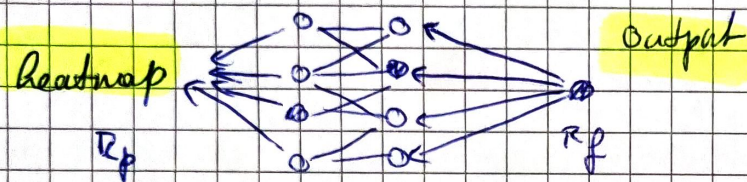
- Visualization of **Critic Maps** during RL
- Observe stability of maps over time

↳ focuses on spatial localization of important regions in the input image for a specific class

Layer-wise Relevance Propagation (LRP)

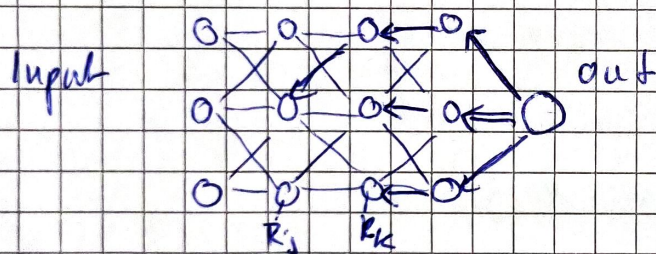


→ Calculates the relevance of each neuron in the input layer of a NN,
↳ Importance of each input feature



→ Utilises layer-structure and propagation mechanism of NN

→ Detailed analysis of the relevance of individual input features through the NN



R_j = relevance of neuron j , z_{jk} = relevance contribution from neuron j to k

$$= a_j \cdot w_{jk}$$

a_j = input activation to neuron j , w_{jk} = weight from neuron j to k

$$R_j = \sum_k \frac{z_{jk}}{\sum_j z_{jk}} R_k$$

• Chat GPT

- general purpose LLM
- prompt
- Strength in Explanation Generation

(+) Natural Language

(+) Versatile in explaining various ML tasks as it is a general-purpose model

(+) Personalization

(+) Interactivity

(+) Multilingual

- Weakness in Explanation Generation

(-) Hallucination / Bias

(-) Bad at reasoning, Generality relies on data set

(-) Hard to ensure the trustworthiness

(-) Inconsistency

(-) Redundant

• Summary

- Hybrid systems makes complementary use of symbolic & ML
- Symbolic rules can be extracted from ML